

# Residual Taxonomy & Bridge Sufficiency for Procedural-Justice Activation Directions

A NeurIPS-style working draft summarizing generated and hand-authored text-benchmark activation evidence. The supported claim is narrow: two different residual classes, two matched repairs, and a negative control blocking universal repair claims.

## Residual classes

2

source pocket + bridge geometry

## Model spaces

4

SmolLM2, Qwen0.5B, Qwen7B, Tiny

## Dissent bridge failures

0

after weighted target bridges

## Claim Boundary

This PDF claims only generated/hand-authored text-benchmark activation diagnostics. It does not claim human behavioral effects, neural alignment, clinical relevance, deployment readiness, or causal steering.

## Accountability

source pocket

## Dissent

bridge geometry

## Withheld

availability gaps

# Two Residual Classes

Failures are not monolithic.

## Generated-reference source pocket

accountability\_after\_harm

Small models invert one recovered generated reference while clean hand-authored accountability controls pass.

Accepted repair:  
strict perturbation  
augmentation.

## Constructed target-bridge geometry pocket

dissent\_after\_mistake

Broad directions score the reference correctly, but target bridges fail when pseudo-side warmth shortcuts are removed.

Accepted repair:  
count-9 target  
bridges with 1:3  
target weighting.

Accepted

Two matched repairs clear all four model spaces.

Rejected

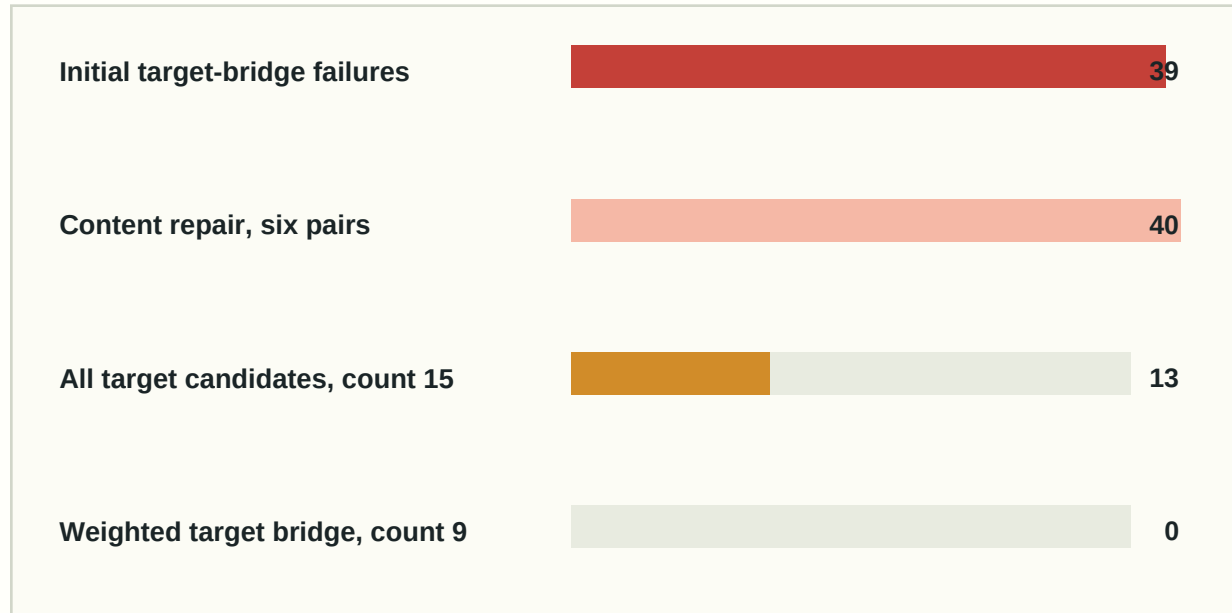
Content-only target repair and universal bridge repair.

Withheld

belonging\_norms and other incomplete availability controls.

# Dissent Bridge Repair Funnel

Rejected paths make the result sharper.



The accepted repair changes bridge construction, not generation. Target bridge pair count stays moderate at 9, but the target-side secondary bridge evidence is repeated 3x while source bridge weights stay at 1:1.

9

target pairs

1:3

target reps

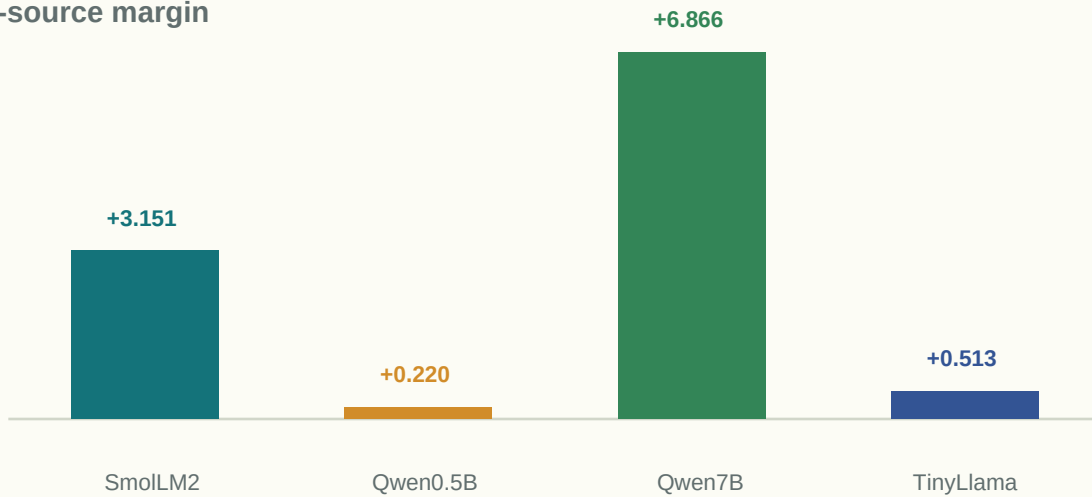
1:1

source reps

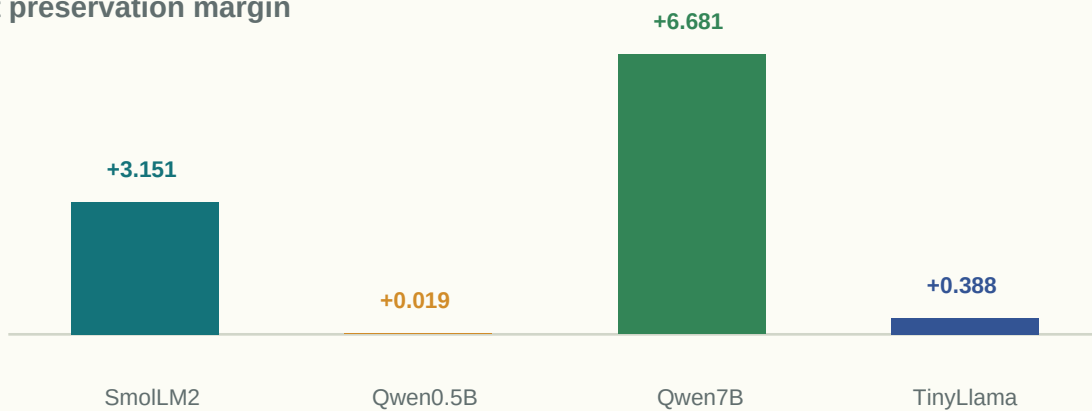
# Accepted Dissent Repair

Fresh slices and preservation both pass.

Fresh-source margin



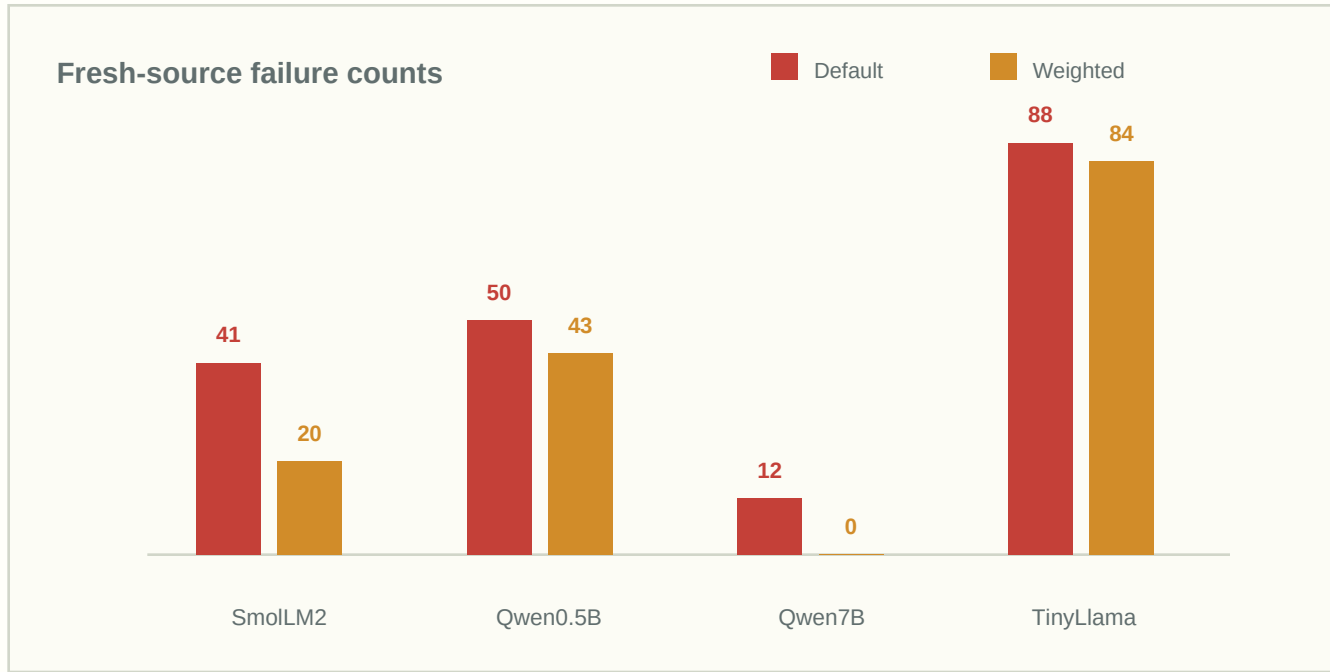
Worst preservation margin



Preservation audit: 32 constructed direction rows, 16 model/evaluation rows, 0 failed pair evaluations, worst margin +0.019.

# Accountability Negative Control

The bridge repair is not universal.



Default count-9

**192** constructed failure rows

Weighted target x3

**147** constructed failure rows

Interpretation: weighted target bridges repair the dissent geometry pocket and preserve its source/control slices. They do not repair the strict accountability generated-reference pocket in three small-model spaces, so the paper claim stays a residual taxonomy rather than a universal recipe.